



8582 - Towards a complete NIR/MIR view of the circumgalactic medium of a brightest cluster galaxy at $z=0.4$

Cycle: 4, Proposal Category: GO

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
MRS				
	1	MRS_BCG_tail_SHOR T_LONG	MIRI Medium Resolution Spectroscopy	(1) MACS1931.8-2635-BCG-TAIL
	2	MRS_Sky_SHORT_L ONG	MIRI Medium Resolution Spectroscopy	(2) DEDICATED-SKY

Folder	Observation	Label	Observing Template	Science Target
NIRSpec				
	5	NIRSpec_BCG_tail	NIRSpec IFU Spectroscopy	(1) MACS1931.8-2635-BCG-TAIL
	6	NIRSpec_Sky	NIRSpec IFU Spectroscopy	(2) DEDICATED-SKY

ABSTRACT

Brightest cluster galaxies (BCGs) of cool-core clusters are uniquely suited for AGN feedback studies. Cycle 2 JWST observations toward one such cool-core BCG have revealed intriguing results: the circumgalactic medium (CGM) has warmer H₂ than in the BCG, suggestive of jet-induced shock heating. Polycyclic aromatic hydrocarbon (PAH) 11.3 μ m emission appears in clumps embedded in the CGM, hinting at an origin in clumpy, dense clouds to survive the destructive environment.

We request NIRSpec+MIRI/MRS IFU observations to complement the Cycle 2 data of this source. While the detections obtained with only 1 of the 3 MRS gratings are promising, full MRS spectral coverage is needed to properly model the continuum and cover the missing spectra features (H₂ S(3), S(7), PAH 6.3, 7.7, 12.6 μ m) to model the gas temperature, excitation and ionization conditions. NIRSpec will detect the PAH 3.3 μ m band and ro-vibrational H₂ lines. The full dataset will allow us to, for the first time, obtain a spatially resolved, contiguous NIRSpec+MRS spectrum over 2-28 micron of a BCG and its surrounding cold CGM, covering $\sim$ 6000 sq kpc. The proposed JWST data will add to a rich ancillary dataset (ALMA, VLT/MUSE, VLA, MeerKAT, Chandra), enabling us to map out the baryon cycle (multiphase gas inflows and outflows), and address fundamental questions on the triggering and quenching of star formation in such environments. We will assess the role of shocks and turbulence in heating the CGM, and understand how molecular gas and PAH survive in harsh conditions in a cluster core. The program will set a new benchmark for the use of PAH as star formation rate indicators, and H₂ lines as molecular gas tracers.

OBSERVING DESCRIPTION

The requested observations are designed to complement the archival Cycle 2 program (3629) obtained with the MRS MEDIUM sub-band

NIRSpec

- G395H/F290LP

- 3x3 mosaic

MIRI/MRS observations towards a spatially extended target

- request is for SHORT and LONG sub-bands, at matching depth to archival observations for MEDIUM sub-band

JWST Proposal 8582 (Created: Thursday, August 21, 2025, 12:00:17PM Eastern Standard Time) - Overview

- 3x2 mosaic
- dedicated sky observation
- simultaneous MIRI imaging (with filters F560W and F100W)
- Note that the simultaneous MIRI imaging over the dedicated sky background will cover the science targets. This is accomplished by restricting the PA range.

Refer to Technical Justification and ETC notebooks for more details

Proposal 8582 - Targets - Towards a complete NIR/MIR view of the circumgalactic medium of a brightest cluster galaxy at z=0.4

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	MACS1931.8-2635-BCG-TAIL	RA: 19 31 49.5580 (292.9564917d) Dec: -26 34 29.29 (-26.57480d) Equinox: J2000	Epoch of Position: 2015.5	
	Comments: Category=Galaxy Description=[Active galactic nuclei, Active galaxies, Brightest cluster galaxies, Radio jets, Starburst galaxies] Extended=YES				
Fixed Targets	(2)	DEDICATED-SKY	RA: 19 31 50.6600 (292.9610833d) Dec: -26 33 18.90 (-26.55525d) Equinox: J2000		
	Comments: Category=Calibration Description=[Telescope/sky background] Extended=YES				

Proposal 8582 - Observation 1 - Towards a complete NIR/MIR view of the circumgalactic medium of a brightest cluster galaxy at z=0.4

Observation	Proposal 8582, Observation 1: MRS_BCG_tail_SHORT_LONG												Thu Aug 21 17:00:17 GMT 2025		
	Diagnostic Status: Warning														
	Observing Template: MIRI Medium Resolution Spectroscopy														
	Background Observations:[MRS_Sky_SHORT_LONG (Obs 2)]														
Diagnostics	(Visit 1:1) Warning (Form): Data Excess over middle threshold														
	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.														
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous				
	(1)	MACS1931.8-2635-BCG-TAIL	RA: 19 31 49.5580 (292.9564917d) Dec: -26 34 29.29 (-26.57480d) Equinox: J2000				Epoch of Position: 2015.5								
	Comments: Category=Galaxy Description=[Active galactic nuclei, Active galaxies, Brightest cluster galaxies, Radio jets, Starburst galaxies] Extended=YES														
Acquisition	#													Target	
	1													NONE	
Template	AcqFilter	Primary Channel				Simultaneous Imaging				Imager Subarray		Grating Wheel Direction			
	F560W	All MRS				YES				FULL		Allow Auto Reorder			
Mosaic	Rows	Columns	Row Overlap %			Column Overlap %			Row shift (deg)		Column shift (deg)		Tile Order		
	2	3	10.0			10.0			20.0		20.0		DEFAULT		
Dithers	#	Dither Type				Optimized For				Direction					
	1	4-Point				EXTENDED SOURCE				NEGATIVE					
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID		
	1		IMAGER	F560W	FASTR1	16	6	1	Dither 1	4	24	1121.116			
	1	SHORT(A)	MRSLONG		FASTR1	104	1	1	Dither 1	4	4	1154.417	218862		
	1	SHORT(A)	MRSSHORT		FASTR1	104	1	1	Dither 1	4	4	1154.417	218862		
	2		IMAGER	F1000W	FASTR1	10	12	1	Dither 1	4	48	1454.121			
	2	LONG(C)	MRSLONG		FASTR1	135	1	1	Dither 1	4	4	1498.522	263381		
	2	LONG(C)	MRSSHORT		FASTR1	135	1	1	Dither 1	4	4	1498.522	263381		

Proposal 8582 - Observation 1 - Towards a complete NIR/MIR view of the circumgalactic medium of a brightest cluster galaxy at z=0.4

Special Requirements	Sequence Observations 1, 2, Non-interruptible
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Proposal 8582 - Observation 2 - Towards a complete NIR/MIR view of the circumgalactic medium of a brightest cluster galaxy at z=0.4

Observation	Proposal 8582, Observation 2: MRS_Sky_SHORT_LONG													Thu Aug 21 17:00:17 GMT 2025
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
	Background Observation For: [MRS_BCG_tail_SHORT_LONG (Obs 1)]													
	Comments: Aperture position angle restricted to ensure that the MIRI imaging covers the science target (BCG+tail); this reduced schedulability													
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous			
	(2)	DEDICATED-SKY	RA: 19 31 50.6600 (292.9610833d) Dec: -26 33 18.90 (-26.55525d) Equinox: J2000											
	Comments: Category=Calibration Description=[Telescope/sky background] Extended=YES													
Acquisition	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel				Simultaneous Imaging				Imager Subarray		Grating Wheel Direction		
		All MRS				YES				FULL		Allow Auto Reorder		
Dithers	#	Dither Type				Optimized For				Direction				
	1	4-Point				EXTENDED SOURCE				NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1		IMAGER	F560W	FASTR1	16	6	1	None	1	6	280.279	218862	
	1	SHORT(A)	MRSLONG		FASTR1	104	1	1	None	1	1	288.604		
	1	SHORT(A)	MRSSHORT		FASTR1	104	1	1	None	1	1	288.604		
	2		IMAGER	F1000W	FASTR1	10	12	1	None	1	12	363.53	263381	
	2	LONG(C)	MRSLONG		FASTR1	135	1	1	None	1	1	374.63		
	2	LONG(C)	MRSSHORT		FASTR1	135	1	1	None	1	1	374.63		

Proposal 8582 - Observation 2 - Towards a complete NIR/MIR view of the circumgalactic medium of a brightest cluster galaxy at z=0.4

Special Requirements	Aperture PA Range 27 to 97 Degrees (V3 27.0 to 97.0) Sequence Observations 1, 2, Non-interruptible
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Proposal 8582 - Observation 5 - Towards a complete NIR/MIR view of the circumgalactic medium of a brightest cluster galaxy at z=0.4

Observation	Proposal 8582, Observation 5: NIRSpec_BCG_tail											Thu Aug 21 17:00:17 GMT 2025
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
	Background Observations:[NIRSpec_Sky (Obs 6)]											
Diagnostics	(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(1)	MACS1931.8-2635-BCG-TAIL	RA: 19 31 49.5580 (292.9564917d) Dec: -26 34 29.29 (-26.57480d) Equinox: J2000				Epoch of Position: 2015.5					
	Comments: Category=Galaxy Description=[Active galactic nuclei, Active galaxies, Brightest cluster galaxies, Radio jets, Starburst galaxies] Extended=YES											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order	
	3	3	10.0		10.0		20.0		20.0		DEFAULT	
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	4-POINT-DITHER										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G395H/F290LP	NRSIRS2RAPID	15	1	false	true	NONE	4	4	933.689	225404

Special Requirements	Sequence Observations 5, 6, Non-interruptible
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Proposal 8582 - Observation 6 - Towards a complete NIR/MIR view of the circumgalactic medium of a brightest cluster galaxy at z=0.4

Observation	Proposal 8582, Observation 6: NIRSpec_Sky Thu Aug 21 17:00:17 GMT 2025											
	Diagnostic Status: Warning Observing Template: NIRSpec IFU Spectroscopy Background Observation For: [NIRSpec_BCG_tail (Obs 5)]											
Diagnostics	(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(2)	DEDICATED-SKY	RA: 19 31 50.6600 (292.9610833d) Dec: -26 33 18.90 (-26.55525d) Equinox: J2000									
Template	<i>Comments:</i> <i>Category=Calibration</i> <i>Description=[Telescope/sky background]</i> <i>Extended=YES</i>											
	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	NONE										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G395H/F290LP	NRSIRS2RAPID	15	1	false	false	NONE	1	1	233.422	
Special Requirements	Sequence Observations 5, 6, Non-interruptible											